Learning Body State from Pressure Sensor Data

Denoising → Representation Learning → Decision-Making, Built on a Synthetic Pilot and Validated on Real PhysioNet PmatData

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What this is: an end-to-end project covering (1) a denoising autoencoder for raw pressure-sensor readings, (2) self-supervised representation learning evaluated under a few-shot labeling budget, with baselines and ablations, and (3) a contextual bandit for deciding which mattress zone to adjust under a hidden user comfort function.

Data: the full pipeline was built and debugged on a programmatic synthetic pressure-map generator with a documented, explicit sensor noise model (code: `src/data.py`), then run **unchanged** on a real, open-access pressure-mat dataset — PmatData (Baran Pouyan et al., IEEE BHI 2017), PhysioNet, 13 human subjects, no login required — to check whether the synthetic-data conclusions actually transfer. Both studies are reported here; see 'Validation on Real Data' for what changed and what didn't.

Reproduce: `pip install -r requirements.txt && bash run_all.sh --with-real` — synthetic study in under a minute on CPU, real-data study a few minutes more (downloads ~107MB from PhysioNet).

Motivation

A pressure-sensing mattress reports a 2D grid of raw taxel readings. Three properties make this a genuinely interesting ML problem rather than a plain classification task. First, the raw signal is noisy — resistive sensor arrays drop cells, spike under cross-talk, and quantize coarsely, so it is worth knowing how to clean the signal and how much that cleaning is actually worth. Second, labels are expensive — a real deployment might realistically get 5–10 labeled posture examples per class, which is exactly the regime where self-supervised pretraining is supposed to pay off. Third, sensing should drive action: the end goal of a smart mattress is a decision (which zone to relieve right now), made under a comfort function that is never directly observable — a contextual bandit problem. The three phases of this project map directly onto these three points.

Pilot Study: Synthetic Data

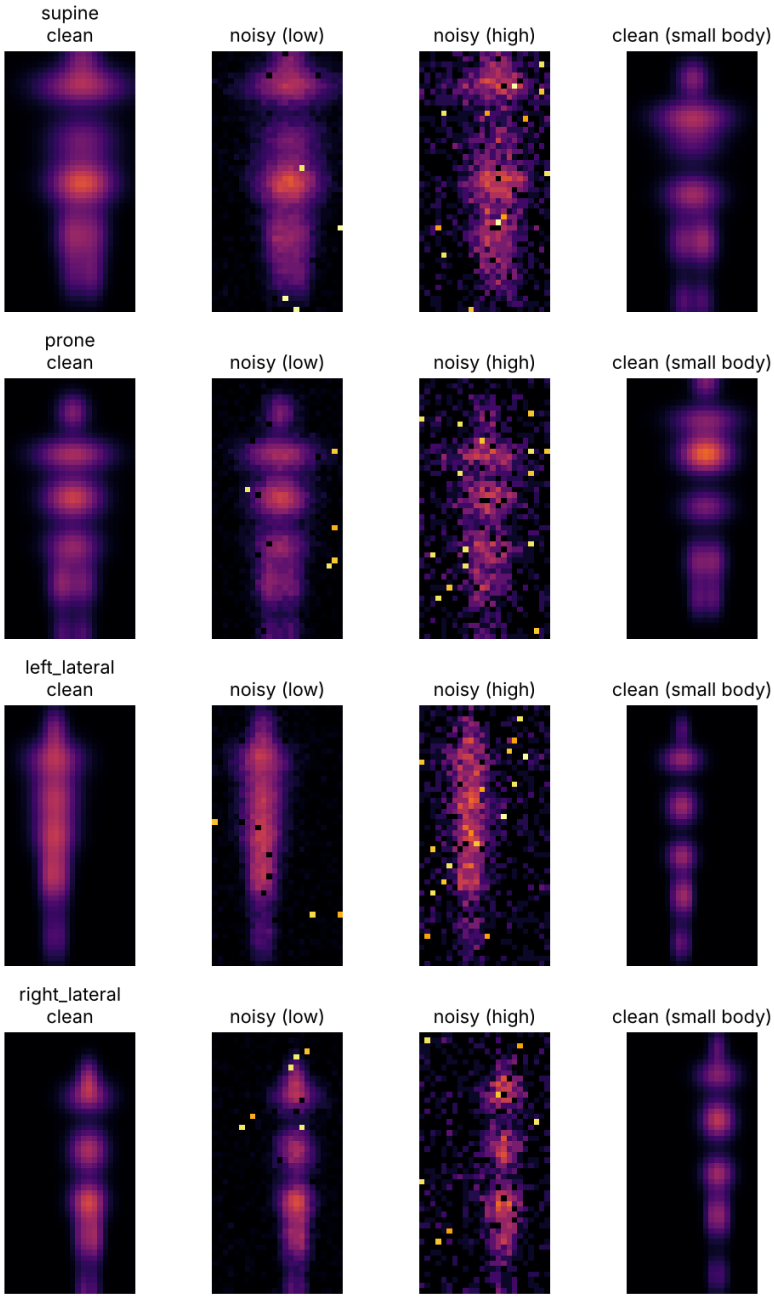
Choice: start synthetic, then validate on real data. The original PMAP dataset is human-subjects data and is request-gated; rather than stall on data access, the pipeline was built and debugged first on a programmatic generator that preserves the structure of the real problem and makes every noise assumption explicit and auditable, then pointed at a real, openly-downloadable pressure-mat dataset with no changes to model or training code once it worked (see 'Validation on Real Data' below). This caught a real bug cheaply, before it could cost real-data debugging time.

The generator produces a 48×24 taxel grid. A body is a sum of anisotropic Gaussian ‘blobs’ (head, shoulders, chest, pelvis, thighs, calves). Posture is the label (`supine, prone, left_lateral, right_lateral`): lateral postures are shifted and narrower (side-lying footprint), while supine vs. prone — the harder pair — differ only in the relative intensity of the chest vs. pelvis blob. Body size (small/medium/large, plus continuous jitter) and a global random placement offset are both generated but withheld from the model as nuisance variables, used later for failure analysis.

Documented sensor noise model:

noisy = quantize(clip(clean + Gaussian(σ) + dead-cell dropout(p_{drop}) + salt spikes(p_{salt}), 0, 1), levels)

with σ, p_{drop}, p_{salt}, and quantization levels scaling together across three severities (low/medium/high, mixed uniformly over the dataset): σ 0.02→0.09, p_{drop} 1%→6%, p_{salt} 0.2%→1.2%, levels 64→16.



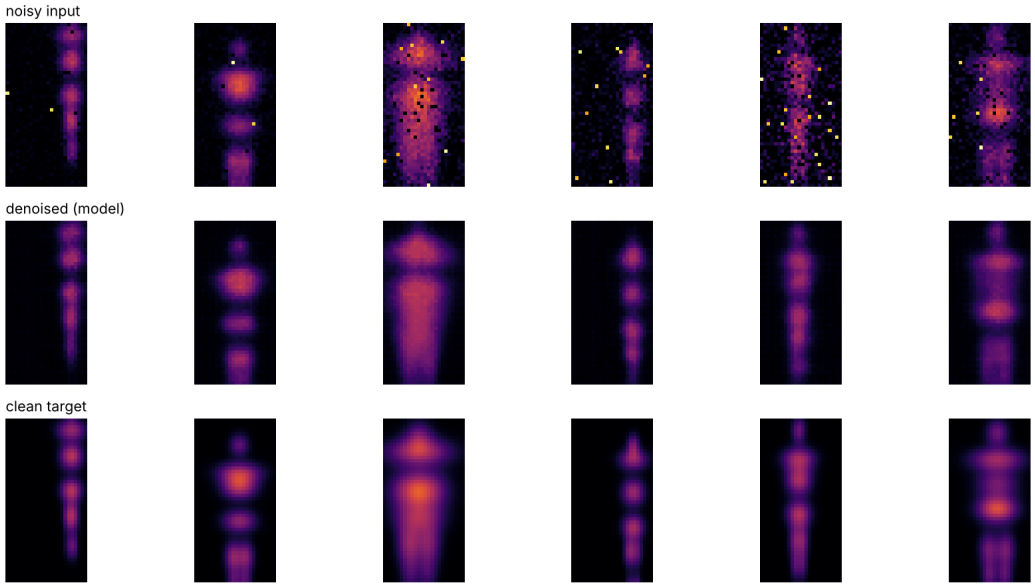
Clean / noisy (low) / noisy (high) / clean-small-body renders for each posture class.

Phase 1 — Denoising (Synthetic Pilot)

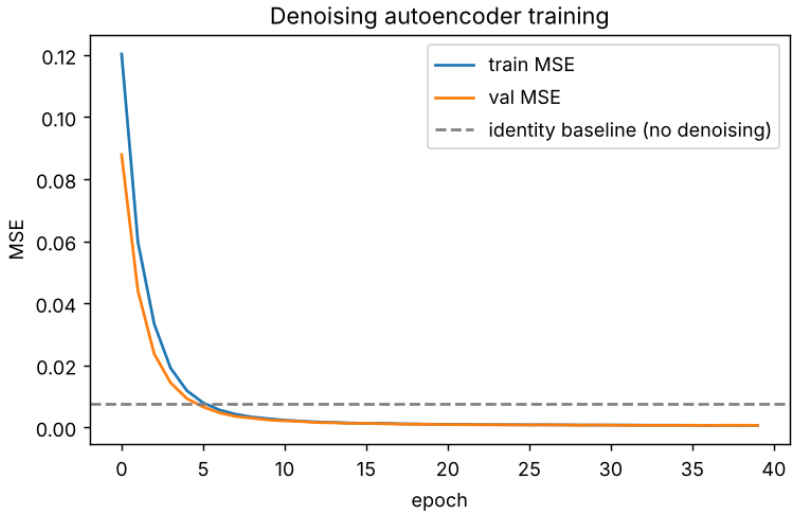
A small conv encoder-decoder (3 stride-2 conv blocks each way, 64-channel bottleneck) trained with MSE to map noisy → clean. 40 epochs, 2000 train / 500 validation pairs, ~14s on CPU.

	MSE	PSNR
Identity baseline (do nothing)	0.00769	21.1 dB
Denoising autoencoder	0.00063	32.0 dB

91.8% reduction in reconstruction MSE vs. doing nothing.



Noisy input / model denoised output / clean target, six held-out validation samples.



Training curve against the identity-baseline MSE floor.

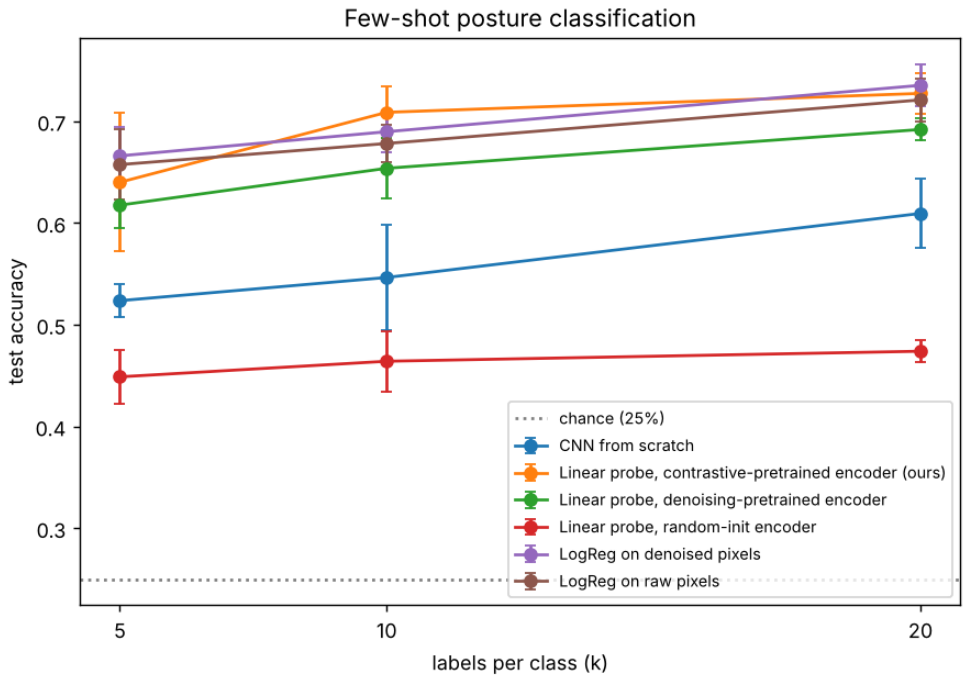
Phase 2 — Representation Learning + Few-Shot Classification (Synthetic Pilot)

The same backbone architecture is pretrained two ways on *unlabeled* data, so only the pretraining signal differs downstream, not the architecture: **contrastive** (SimCLR-style NT-Xent, temperature 0.3, augmentations = noise/translation/dropout/intensity-scale — deliberately no horizontal flip, since flipping would swap left/right lateral labels), and **denoising** (the Phase 1 autoencoder's backbone, reused directly).

Evaluated for k ∈ {5, 10, 20} labels/class (4 classes), averaged over 8 random label draws, against a held-out test set, six methods:

method	k=5	k=10	k=20
Linear probe, random-init encoder (floor)	44.9%	46.4%	47.4%
CNN trained from scratch (k-shot only)	52.4%	54.6%	60.9%
Linear probe, denoising-pretrained encoder	61.7%	65.4%	69.2%
LogReg on raw noisy pixels	65.7%	67.8%	72.1%
LogReg on denoised pixels	66.6%	69.0%	73.5%
Linear probe, contrastive-pretrained encoder (ours)	64.0%	70.9%	72.7%

(chance = 25%; full numbers with std in reports/results/fewshot_summary.json)



Few-shot test accuracy vs. labels per class, mean ± std over 8 draws.

Step 3 — Evaluating Like a Researcher

Ablations

- **Does pretraining help, and by how much?** Yes, decisively, against the only fair control (same architecture, random weights): the random-init linear probe sits at 45–47% regardless of k, while either pretraining method lifts it to 62–73% — a +17 to +26 point jump.
- **Does denoising help downstream accuracy?** Yes, modestly and consistently: LogReg on denoised pixels beats LogReg on raw noisy pixels at every k (+0.9, +1.2, +1.4 points). A real but small effect.
- **Does the CNN-from-scratch baseline justify its cost?** No — it is the worst non-floor method at every k, confirming that training a full conv net end-to-end on 20–80 total labels overfits badly compared to fitting a linear head on good frozen features.

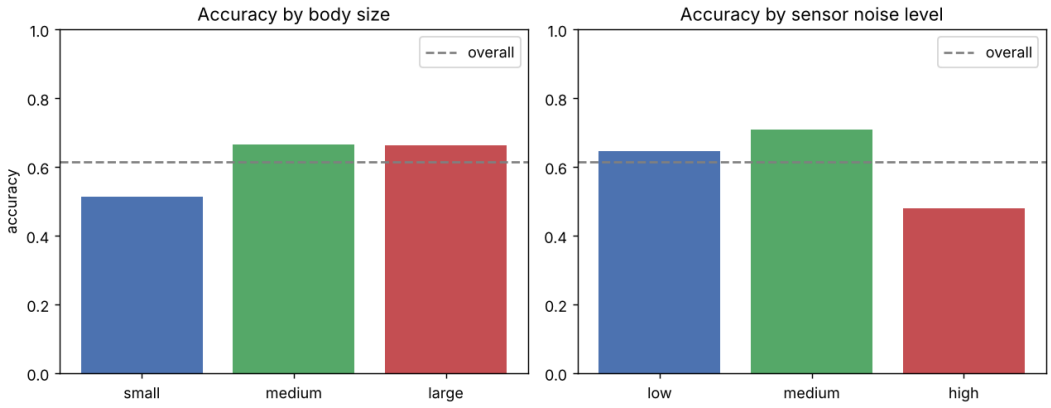
Honest surprise: raw pixels are a strong baseline

The first version of this project used perfectly-centered bodies, and raw-pixel logistic regression hit **99%+ accuracy with only 10 labels/class** — beating every pretrained representation by 20+ points. That is not a success story, it is a sign the synthetic task was too easy: with no nuisance variation, posture differences are a handful of cleanly linearly-separable pixels, and 1152-dimensional logistic regression with 5–20 samples per class will happily overfit a perfect linear boundary in that regime (classic n■d). The fix, documented in `src/data.py`, was to add a **global random body placement offset per sample**, independent of posture — the subject is not perfectly centered on the mattress, which is true of real sensor data. That single change dropped raw-pixel LogReg from 99%→~70% and let the representation-learning story actually show up. **Lesson:** a synthetic benchmark that makes your proposed method look great by a huge margin is itself a result worth being suspicious of.

Even after the fix, raw-pixel LogReg stays *competitive* with the pretrained linear probes rather than clearly losing — on a 48×24, 4-class problem a 1152-dim linear boundary is a genuinely strong baseline. We report this rather than cherry-picking a setting that flatters the pretrained encoder.

Failure analysis

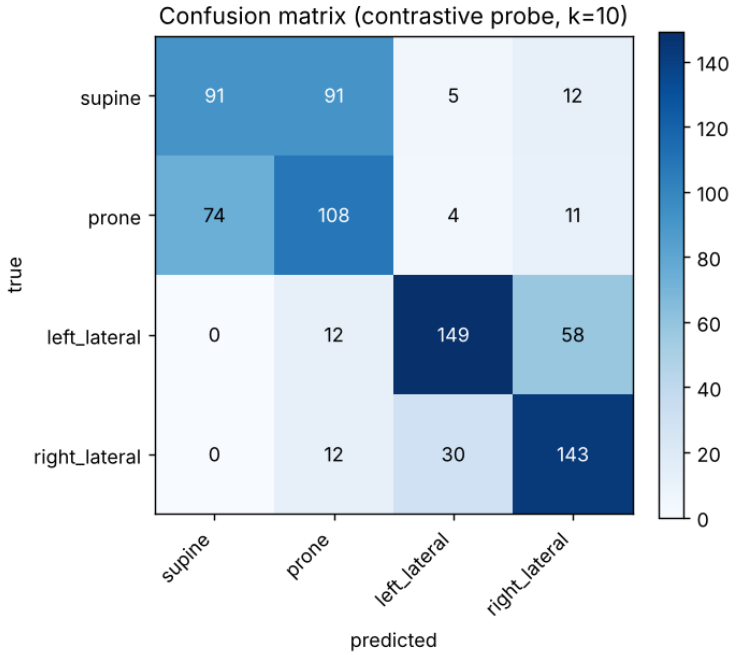
Contrastive probe, k=10, a single representative label draw, evaluated on the full 800-sample held-out test set (overall accuracy 61.4%):



Accuracy broken down by nuisance variables never given to the model: body size and sensor noise level.

• **Body size:** small bodies are the clear weak point (51.3% vs. 66.5% / 66.4% for medium/large) — smaller blobs produce weaker, more noise-fragile pressure signatures relative to a fixed sensor-noise magnitude.

• **Sensor noise level:** accuracy drops sharply at high noise (48.1%) vs. low/medium (64.5% / 70.9%) — a reminder that ‘works on clean sensors’ and ‘works in the field’ are different claims.

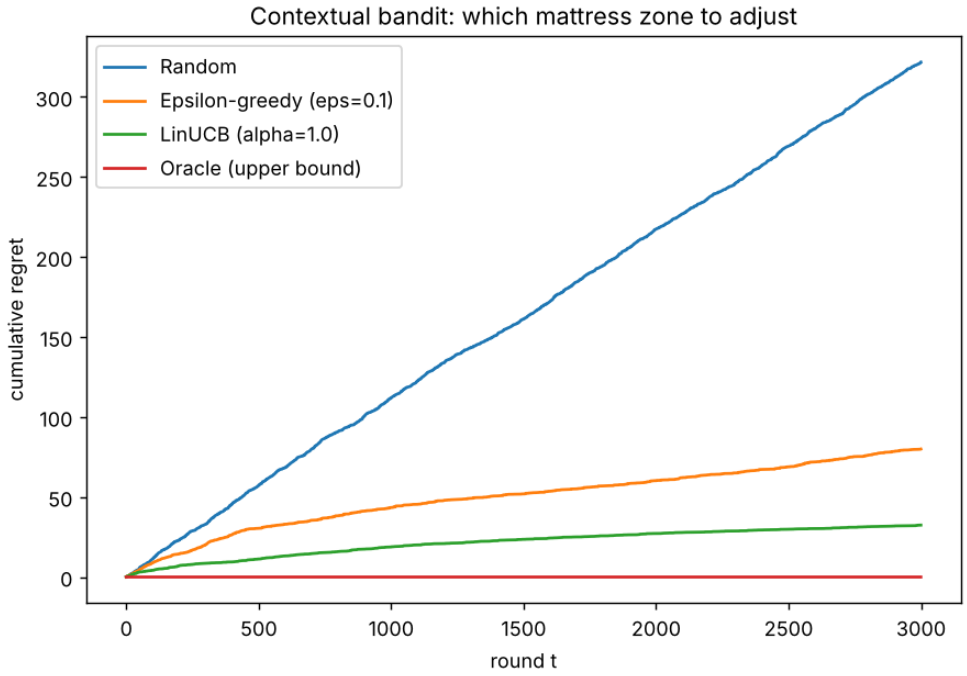


Supine and prone are confused with each other constantly, essentially never with either lateral posture; left/right lateral confuse each other far less. This matches the generative design exactly: lateral-vs-not is a gross, noise-robust spatial signal, while supine-vs-prone is a subtle chest/pelvis intensity ratio that noise and small body size both directly corrupt.

Phase 3 (Stretch) — Contextual Bandit for Zone Adjustment (Synthetic Pilot)

The mattress is split into 5 zones (head, shoulders, lower-back/hips, thighs, calves). At each round the agent observes per-zone mean pressure as context and picks one zone to relieve. A simulated user has a hidden, fixed per-zone comfort-sensitivity vector θ (unknown to the agent): reward(zone z) = $\theta_z \cdot \text{pressure}_z$ + noise — a disjoint linear contextual bandit. Policies: random, epsilon-greedy (per-arm ridge regression, ε=0.1), LinUCB (disjoint, Li et al. 2010, α=1.0), and an oracle that knows θ exactly. T=3000 rounds.

policy	final cumulative regret	mean reward (last 500 rounds)
Random	321.8	0.139
Epsilon-greedy	80.0	0.222
LinUCB	32.5	0.239
Oracle (upper bound)	0.0	0.244



LinUCB's context-aware confidence bound closes the gap to oracle far faster than epsilon-greedy's undirected exploration. Regret is visibly sublinear (concave) for both learning policies and linear for random, exactly as theory predicts.

Validation on Real Data (PmatData)

Source: PmatData, “A Pressure Map Dataset for In-bed Posture Classification” (Baran Pouyan et al., IEEE BHI 2017), QoLT Lab, UT Dallas. Open access on PhysioNet under the Open Data Commons Attribution License v1.0 — **no login or credentialing required**, downloaded directly (`scripts/08_download_real_data.sh`). Experiment I: 13 subjects, a real Vista Medical FSA SoftFlex 2048 mat (32×64 taxels), 17 labeled positions per subject, ~20,000 real pressure frames in total.

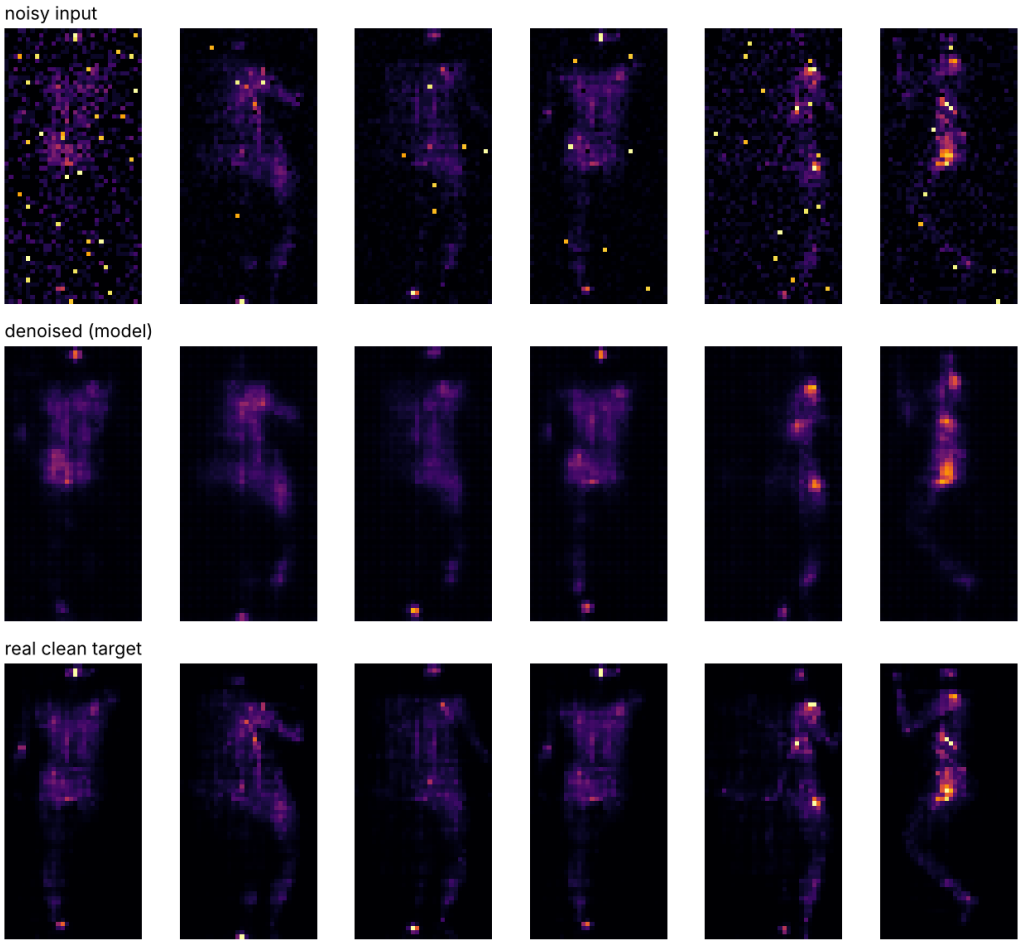
Labels: transcribed directly from the dataset's own position table. The real dataset has **no prone posture** recorded, so this study uses 3 classes (supine/left/right) instead of the synthetic study's 4. Bed inclination and body-roll wedges/fetal position are recorded and used as a real nuisance-variant label, the direct analogue of synthetic `body_size`.

Split: 3 of 13 subjects held out entirely — none of their frames appear in pretraining, denoising, or few-shot label selection. Test accuracy below measures generalization to an **unseen person's body**, stricter than the synthetic study's frame-level split.

Denoising on real data

	MSE	PSNR
Identity baseline	0.00770	21.1 dB
Denoising autoencoder (real data)	0.00204	26.9 dB

73.4% MSE reduction — working, but weaker than the synthetic pilot's 91.8%/32.0dB. Real pressure maps have fine, sparse, high-frequency structure (fingers, heel points, shoulder blades) a small bottleneck smooths over; synthetic Gaussian blobs are smooth by construction and easier to reconstruct.



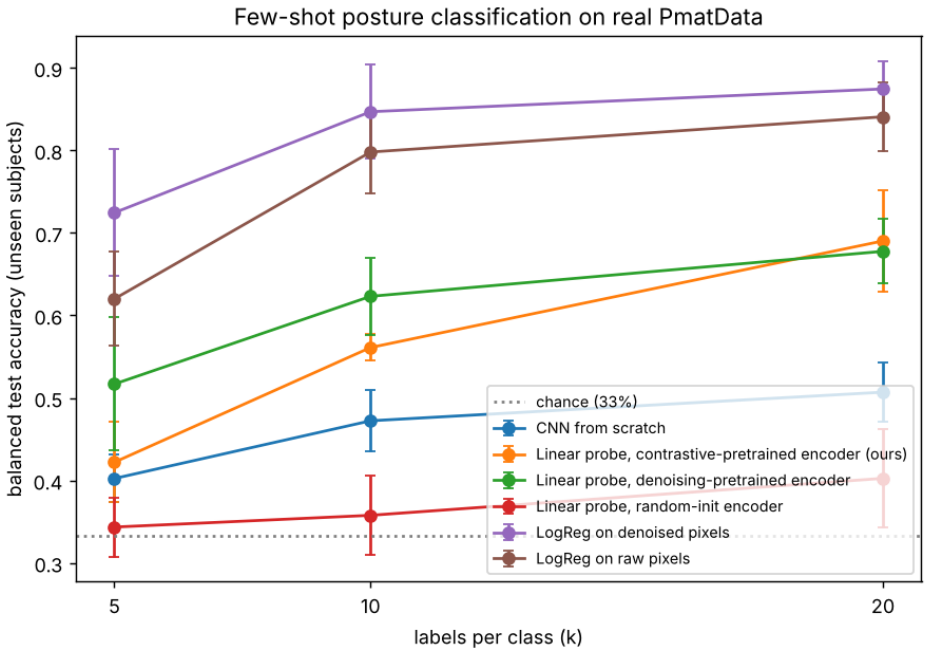
Real noisy input / model denoised output / real clean target, held-out validation samples.

Few-shot posture classification on real data (unseen subjects)

Same six methods and k ∈ {5,10,20} protocol, evaluated on 3 held-out subjects. Real posture classes are imbalanced (supine covers 9 of 17 positions), so **balanced accuracy** (macro-recall) is reported:

method	k=5	k=10	k=20
Linear probe, random-init encoder (floor)	34.3%	35.8%	40.3%
CNN trained from scratch	40.2%	47.2%	50.7%
Linear probe, contrastive-pretrained encoder	42.2%	56.1%	69.0%
Linear probe, denoising-pretrained encoder	51.7%	62.3%	67.8%
LogReg on raw pixels	62.0%	79.8%	84.1%
LogReg on denoised pixels	72.4%	84.7%	87.4%

(balanced accuracy; chance = 33%; full numbers in reports/results/real_fewshot_summary.json)



Balanced few-shot test accuracy on unseen subjects, mean ± std over 8 draws.

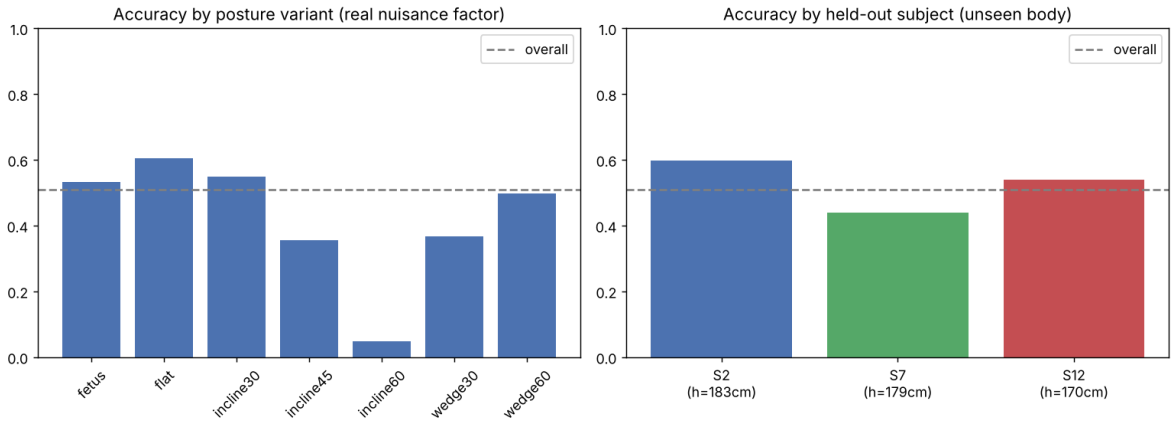
This is a materially different result from the synthetic pilot, reported as-is rather than adjusted to match the earlier narrative. On real data, raw-pixel logistic regression — especially on denoised pixels — dominates by a wide margin at every k, more decisively than on synthetic data. This is not the earlier bug repeating: real supine/left/right postures produce dramatically different, highly reproducible pixel silhouettes (a side-lying body rotates and narrows the whole footprint), so a high-dimensional linear classifier with even 10–20 examples exploits that structure very effectively — genuine task structure this time, not a generator artifact. These three coarse classes are simply easier to separate by gross shape than the synthetic pilot's hardest pair (supine vs. prone, a subtle intensity-ratio cue) — and prone isn't in this real dataset to test against.

The pretraining ablation still holds: both pretrained linear probes beat the random-init floor by a wide margin (+8 to +29 balanced-accuracy points) at every k, confirming pretraining learns something real about body pressure

structure even on a task where it isn't the best available method.

Failure analysis on real data

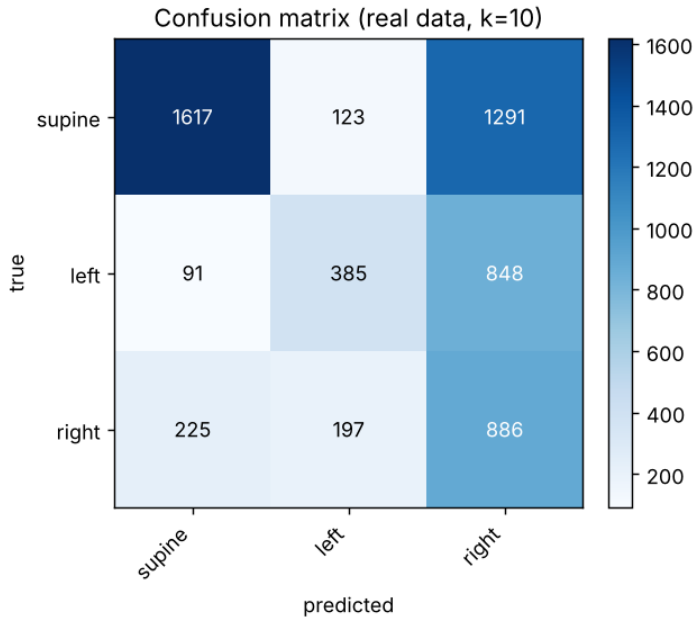
Contrastive linear probe, k=10, one representative label draw, all 3 held-out subjects (overall accuracy 51.0%, balanced 50.1%):



Accuracy by real nuisance variant (bed incline/wedge/fetus) and by held-out subject.

• **Bed inclination is the standout failure mode, with a clean dose-response curve:** flat supine 60.4% → 30° 54.9% → 45° 35.6% → 60° **4.9%** — worse than chance. The model was trained almost entirely on flat postures, so a steeply inclined supine posture, which genuinely redistributes pressure toward the hips/lower back, is essentially out-of-distribution. The real-data analogue of the synthetic study's 'small body size' finding: a real physical nuisance variable the model was never taught to be invariant to.

• **Body-roll wedges** (36.8% at wedge30) also hurt noticeably more than the flat baseline (60.4%), while fetal position (53.3%) is closer to normal.

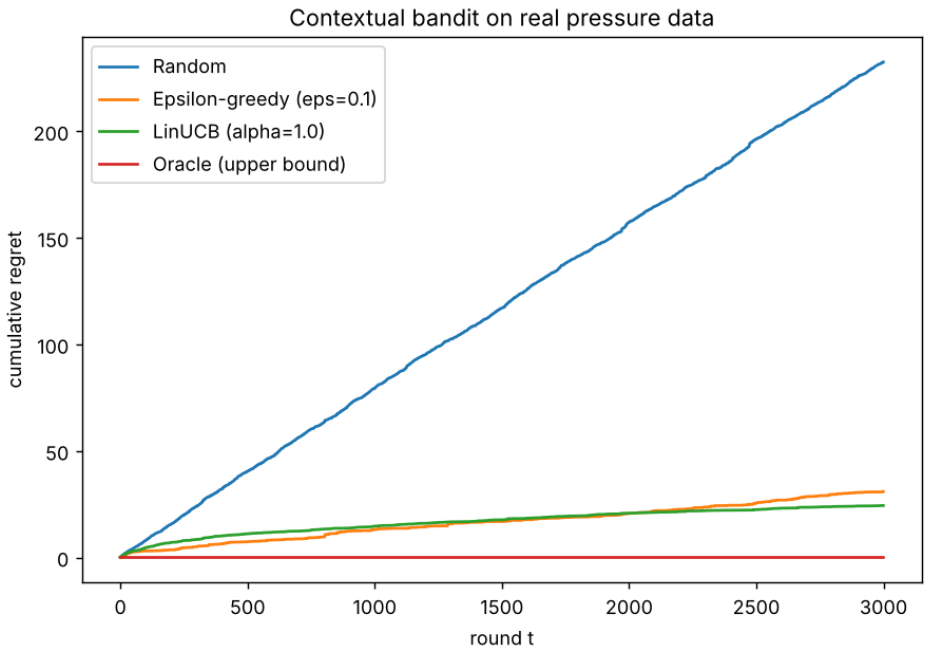


This single k=10 draw's confusion matrix shows a classifier biased toward predicting 'right' — a few-shot **instability** symptom (a linear probe fit on 10 examples/class in a 64-dim space can land on a biased boundary by chance on any single draw), not a stable property of the method. The main few-shot table (averaged over 8 draws) is the number to trust; this single-draw matrix is shown because it illustrates that instability honestly.

Contextual bandit on real data

Identical bandit simulation to the synthetic pilot (5 zones, disjoint linear reward, hidden per-zone sensitivity θ), with the context stream replaced by real PmatData frames — no other change.

policy	final cumulative regret	mean reward (last 500)	% of oracle reward
Random	232.5	0.041	36.3%
Epsilon-greedy	30.9	0.103	90.3%
LinUCB	24.3	0.110	96.6%
Oracle (upper bound)	0.0	0.114	100.0%



Identical qualitative story to the synthetic pilot: LinUCB reaches within 3.4% of oracle reward, clearly beating epsilon-greedy and random. This part of the pipeline was context-agnostic by design and transfers to real data without modification, as expected.

What changed between synthetic and real, in one paragraph

Denoising works on both, modestly weaker on real data's finer detail. Pretraining reliably beats random-init features on both. What doesn't transfer is which method wins outright: on the harder synthetic task (subtle supine/prone cue, deliberate placement jitter), contrastive pretraining is competitive-to-best; on the real, coarser 3-class task, simple logistic regression on (denoised) pixels wins decisively, because real supine/left/right postures are that linearly separable in pixel space. Neither result is wrong — they answer different questions about different tasks, and reporting both rather than only the one that flatters representation learning is the point of doing this validation at all.

Limitations

• **The real-data study still doesn't cover everything a deployed system would face.** PmatData is 13 subjects on one mat model in a lab setting — it captures real anatomical variation and genuine sensor noise/sparsity, but still has no prone posture, no long-duration sessions, and no sensor drift across weeks. 'Validated on real data' means real recordings, not a real deployment.

• **Representation learning's advantage is real but task-dependent, not dominant — more so than expected.** On synthetic data, contrastive pretraining was competitive-to-best against raw pixels; on real data, raw-pixel LogReg wins decisively, because real supine/left/right postures are extremely linearly separable by gross silhouette. The honest claim across both studies: pretrained features reliably beat a random-init encoder; whether they beat a strong linear baseline depends on how fine-grained the distinction is.

• **The bandit's comfort function is linear and stationary,** in both studies, and was only ever validated on simulated reward, never a real comfort signal (none exists in PmatData).

• **Few-shot linear probes are unstable on a single label draw,** confirmed directly in the real-data failure analysis. The averaged, multi-draw few-shot tables are the trustworthy numbers; a single deployed few-shot classifier should expect similar draw-to-draw variance.

• **No real-world sensor drift or calibration across sessions was modeled** in either study.

Future Work

• **Done in this iteration:** validated the synthetic-data conclusions against a real pressure-mat dataset. Result: pretraining-beats-random-init transferred cleanly; 'pretraining beats simple baselines outright' did not — a finding worth having, not a failure to paper over.

• Test whether representation learning's advantage reappears on a harder real discrimination task — PmatData's own bed-incline/body-roll variants (used here only as nuisance labels) could be turned into finer-grained labels closer in spirit to the synthetic study's subtle supine/prone cue.

• Try a denoising + contrastive joint pretraining objective to see if the two signals are complementary rather than alternatives.

• Extend the bandit's reward model to be non-stationary and to richer action spaces; validating against even simulated-but-individualized human comfort ratings would be stronger than the current fully-synthetic reward.

• Bring in PmatData's Experiment II (a second, different mat model) as a cross-hardware generalization test: train on Experiment I, evaluate on Experiment II unchanged.